Lesson 07: Switched outlets

Switched-outlet model | 12 V AC

Name: [redacted] Class: _____ Date: 04/11/26

Build and change circuits with the supply disconnected. Ask for a teacher check before testing.

1. Why did our circuits fail?

Two possible causes of our failed circuits: The wires weren't working properly

2. Household conductors: A, N and E

Match each conductor to a function: 1 = normal return path; 2 = protective fault path; 3 = supply conductor.

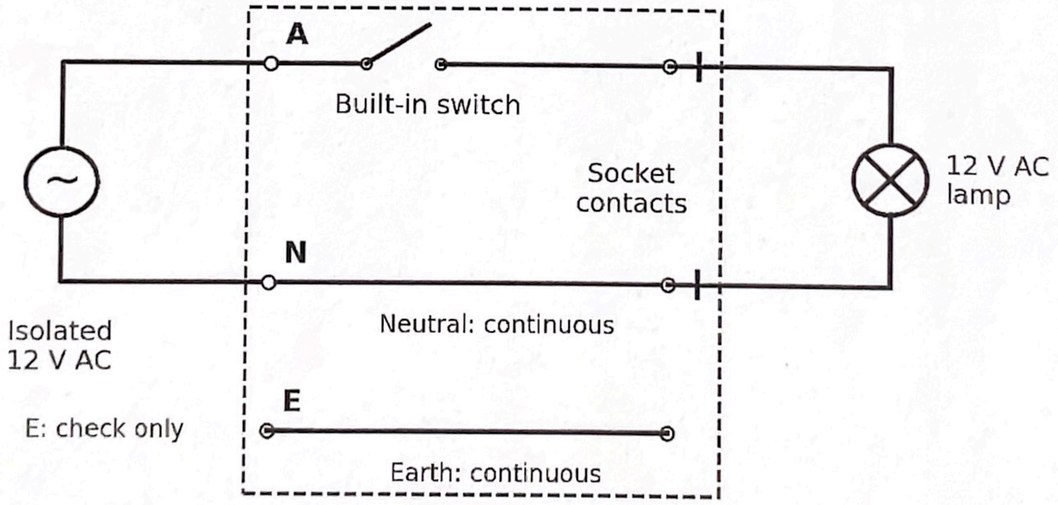
A (active/live): 3 N (neutral): 1 E (protective earth): 2

These are household roles. Today's supply has two isolated 12 V AC terminals.

3. Connect and test the switched outlet

Connect the supply cable to the outlet's labelled terminals using the checked example. Its switch-to-socket connection is already built in. Test with a 12 V AC lamp.

ONE SWITCHED SOCKET: INTERNAL PATHS



A/N/E name the fixture terminals, not the isolated supply outputs. Check E continuity separately. Common is a switch-contact label, not neutral.

Teacher check: ✓ Open: lamp ✓ Closed: lamp ✓

The built-in switch interrupts: ✓

4. Exit check

One connection-quality check: _____

Why switch active rather than only neutral? _____

Lesson 07: Switched outlets

Switched-outlet model | 12 V AC

Name: [redacted] Class: Science Date: 4/11/26

Build and change circuits with the supply disconnected. Ask for a teacher check before testing.

1. Why did our circuits fail?

Two possible causes of our failed circuits: The wires are broken inside Chord

2. Household conductors: A, N and E

the joinings aren't ~~from~~ formed properly crocodile clips

Match each conductor to a function: 1 = normal return path; 2 = protective fault path; 3 = supply conductor.

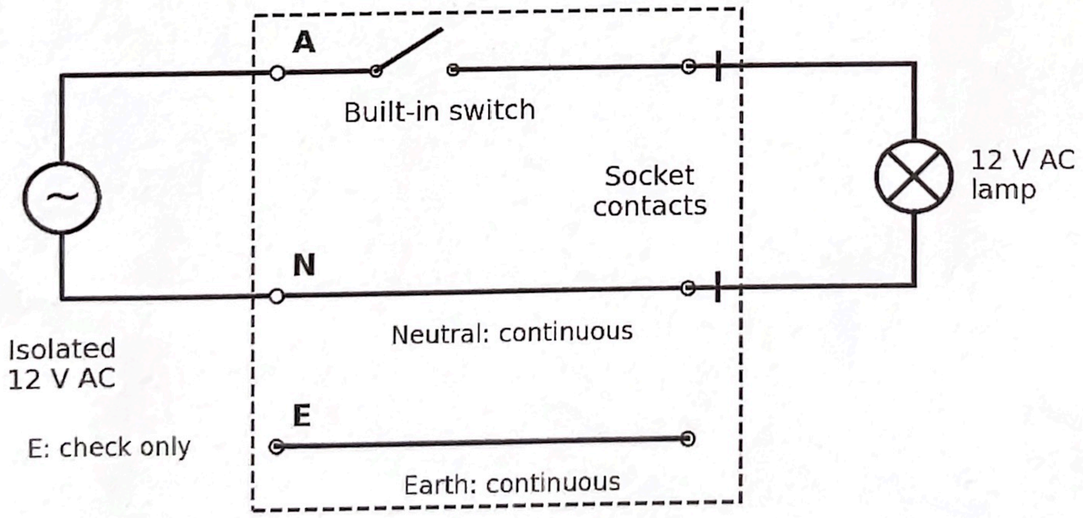
A (active/live): 3 N (neutral): 1 E (protective earth): 2

These are household roles. Today's supply has two isolated 12 V AC terminals.

3. Connect and test the switched outlet

Connect the supply cable to the outlet's labelled terminals using the checked example. Its switch-to-socket connection is already built in. Test with a 12 V AC lamp.

ONE SWITCHED SOCKET: INTERNAL PATHS



A/N/E name the fixture terminals, not the isolated supply outputs. Check E continuity separately. Common is a switch-contact label, not neutral.

Teacher check: ✓ Open: lamp ✓ Closed: lamp ✓

The built-in switch interrupts: ✓

4. Exit check

One connection-quality check: _____

Why switch active rather than only neutral? _____

Lesson 07: Switched outlets

Switched-outlet model | 12 V AC

Name [redacted] Class: Science Date: 4/11/26

Build and change circuits with the supply disconnected. Ask for a teacher check before testing.

1. Why did our circuits fail?

Two possible causes of our failed circuits: The wires are broken inside the chord
The crocodile clips aren't connected

2. Household conductors: A, N and E

Match each conductor to a function: 1 = normal return path; 2 = protective fault path; 3 = supply conductor.

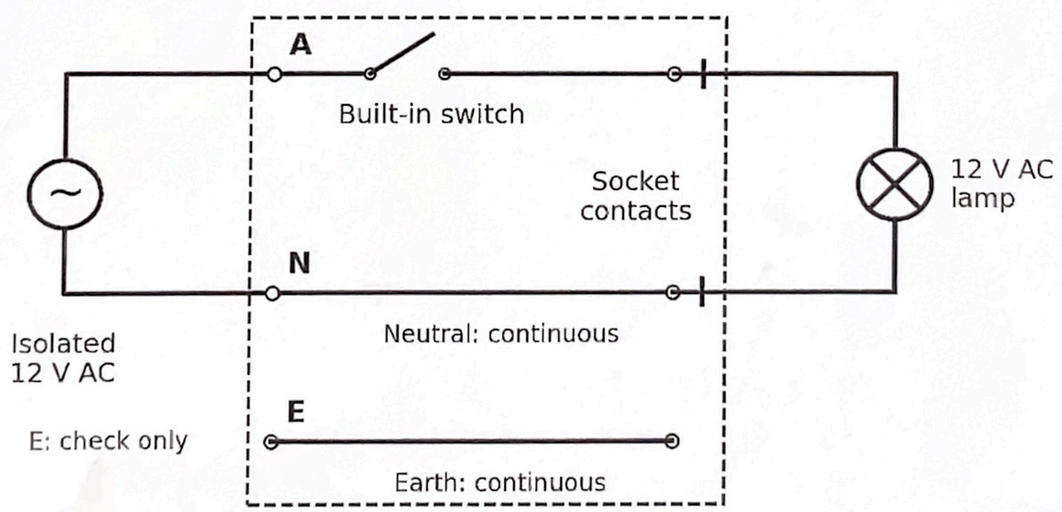
A (active/live): 3 N (neutral): 1 E (protective earth): 2

These are household roles. Today's supply has two isolated 12 V AC terminals.

3. Connect and test the switched outlet

Connect the supply cable to the outlet's labelled terminals using the checked example. Its switch-to-socket connection is already built in. Test with a 12 V AC lamp.

ONE SWITCHED SOCKET: INTERNAL PATHS



A/N/E name the fixture terminals, not the isolated supply outputs. Check E continuity separately. Common is a switch-contact label, not neutral.

Teacher check: ✓ Open: lamp ✓ Closed: lamp ✓

The built-in switch interrupts: ✓

4. Exit check

One connection-quality check: _____

Why switch active rather than only neutral? _____

Lesson 07: Switched outlets

Switched-outlet model | 12 V AC

Name: [redacted] Class: 10A Date: 4/10/26

Build and change circuits with the supply disconnected. Ask for a teacher check before testing.

1. Why did our circuits fail?

Two possible causes of our failed circuits: Broken wires, incorrect placement

2. Household conductors: A, N and E

Match each conductor to a function: 1 = normal return path; 2 = protective fault path; 3 = supply conductor.

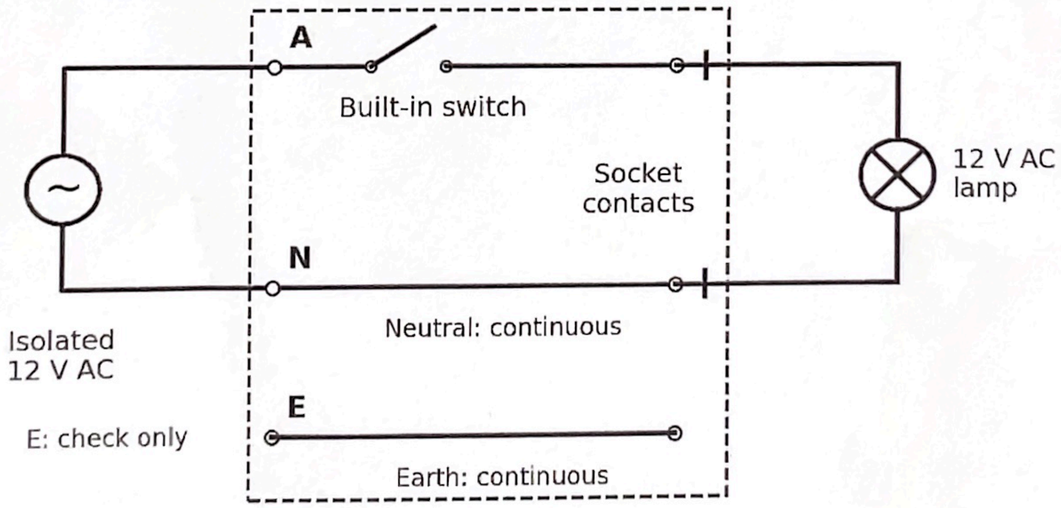
A (active/live): 3 N (neutral): 1 E (protective earth): 2

These are household roles. Today's supply has two isolated 12 V AC terminals.

3. Connect and test the switched outlet

Connect the supply cable to the outlet's labelled terminals using the checked example. Its switch-to-socket connection is already built in. Test with a 12 V AC lamp.

ONE SWITCHED SOCKET: INTERNAL PATHS



A/N/E name the fixture terminals, not the isolated supply outputs. Check E continuity separately. Common is a switch-contact label, not neutral.

Teacher check: ✓ Open: lamp ✓ Closed: lamp ✓

The built-in switch interrupts: ✓

4. Exit check

One connection-quality check: _____

Why switch active rather than only neutral? _____

Lesson 07: Switched outlets

Switched-outlet model | 12 V AC

Name: [redacted] Class: Science Date: 4/11/26

Build and change circuits with the supply disconnected. Ask for a teacher check before testing.

1. Why did our circuits fail?

Two possible causes of our failed circuits: wires are broken flowing improper

2. Household conductors: A, N and E

Match each conductor to a function: 1 = normal return path; 2 = protective fault path; 3 = supply conductor.

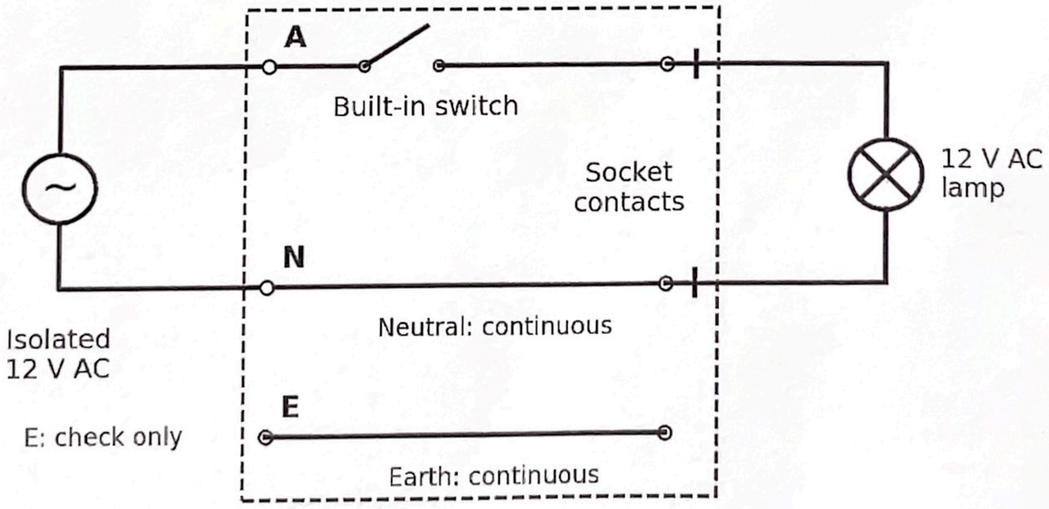
A (active/live): 2 N (neutral): 1 E (protective earth): 3

These are household roles. Today's supply has two isolated 12 V AC terminals.

3. Connect and test the switched outlet

Connect the supply cable to the outlet's labelled terminals using the checked example. Its switch-to-socket connection is already built in. Test with a 12 V AC lamp.

ONE SWITCHED SOCKET: INTERNAL PATHS



A/N/E name the fixture terminals, not the isolated supply outputs. Check E continuity separately. Common is a switch-contact label, not neutral.

Teacher check: ✓ Open: lamp ✓ Closed: lamp ✓

The built-in switch interrupts: ✓

4. Exit check

One connection-quality check: _____

Why switch active rather than only neutral? _____

Lesson 07: Switched outlets

Switched-outlet model | 12 V AC

Name [redacted] Class: 10G Date: 4/11/26

Build and change circuits with the supply disconnected. Ask for a teacher check before testing.

1. Why did our circuits fail?

Two possible causes of our failed circuits: Broken wires, Broken clips

2. Household conductors: A, N and E

Match each conductor to a function: 1 = normal return path; 2 = protective fault path; 3 = supply conductor.

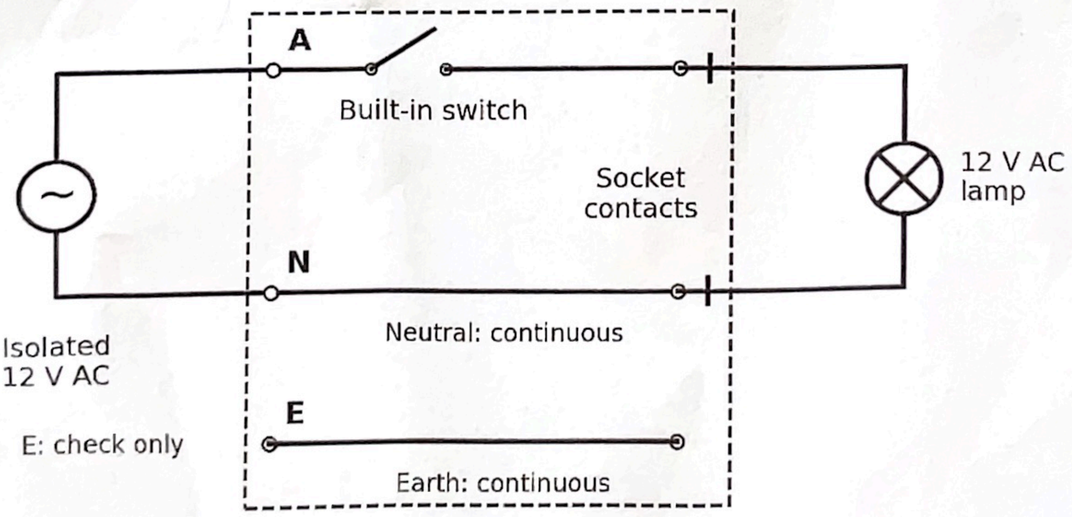
A (active/live): 3 N (neutral): 1 E (protective earth): 2

These are household roles. Today's supply has two isolated 12 V AC terminals.

3. Connect and test the switched outlet

Connect the supply cable to the outlet's labelled terminals using the checked example. Its switch-to-socket connection is already built in. Test with a 12 V AC lamp.

ONE SWITCHED SOCKET: INTERNAL PATHS



A/N/E name the fixture terminals, not the isolated supply outputs. Check E continuity separately. Common is a switch-contact label, not neutral.

Teacher check: ✓ Open: lamp ✓ Closed: lamp ✓

The built-in switch interrupts: ✓

4. Exit check

One connection-quality check: _____

Why switch active rather than only neutral? _____